

Mukundkumar Patel

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EDUCATION:

Montclair State University (GPA: 4.0/4.0)

September 2024 - May 2026

Master of Science: Data Science

CVM University (GPA: 3.58/4.0)

June 2020 - March 2024

Bachelor of Engineering: Information Technology

TECHNICAL SKILLS:

- Programming languages: Python, SQL, R, DAX, Power Query
- Tools: MS Power BI, MS Excel (VLOOKUP, Pivot Tables), ArcGIS, Tableau, MS PowerPoint
- Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, Folium, Plotly, Google Gemini, Anthropic API
- Database and Cloud: MySQL, Oracle, SQLite, Data Lake, PostgreSQL, MS Access

WORK EXPERIENCE:

Data Analyst Intern (MTA - Metropolitan Transportation Authority)

October 2025 - May 2026

- **Automated data processing** and data validation of vendor-provided data for 8,500+ events **using Python and Office Scripts**, identifying and correcting inaccuracies to improve overall data quality.
- Managed and improved master data for 1,400+ bus routes using SQL and Excel; performed **ArcGIS-based spatial analysis** to identify route overlaps and improve operational efficiency.
- Designed a Python computer vision pipeline to detect bus lane and bus stop violations from 900+ traffic cameras, achieving 92% detection accuracy; integrated results with **Power BI** dashboards to visualize violation trends for accurate ticket issuance.
- Built a Python **ETL workflow** to process driver trip logs, calculating pull-in/pull-out counts per depot at 30-minute intervals to optimize fleet planning, maintenance schedules, and backup coordination.

Graduate Research Assistant (Montclair State University)

August 2025 - May 2026

- Standardized and governed institution-wide student master data (8,000+ records) using SQL and spreadsheets for federal compliance, institutional research, and governmental regulatory reporting.
- Created **Tableau** dashboards to analyze 5+ years of student data across enrollment, financial aid, and academic outcomes; presented KPI trends and performance reporting insights to stakeholders for leadership review.

Data Analyst (Montclair State University)

April 2025 - August 2025

- Streamlined the data cleaning process, **reducing processing time by 30%**, and structured raw data into categorized community partners, focus areas, and population segments for targeted impact analysis.
- Developed an interactive **Python map** and **Power BI** dashboards to analyze 150+ community investment activities, visualize top-performing initiatives, track month-wise trends, and identify key regions for targeted investment.

PROJECT DETAILS:

Sustainable Shipping Route Optimizer ([GitHub](#)) ([Dashboard](#))

April 2026

- Designed logistics optimization platform using **Python**, **Google Gemini AI**, and **Streamlit** that generates carbon-optimized shipping routes across 60+ US cities by analyzing diesel truck, electric vehicle, and rail freight options using **EPA SmartWay carbon calculation** methodology and multi-objective optimization algorithms.
- Implemented **RESTful API** integration (**OpenRouteService**, **OpenWeatherMap**), **SQLite** database for intelligent route caching, and natural language processing interface enabling conversational route queries with real-time cost, delivery time, and carbon footprint analysis for data-driven logistics decision-making.

Sales and Demand Forecasting ([GitHub](#)) ([Dashboard](#))

November 2023

- Designed and fine-tuned ARIMA and SARIMA models in Python to **forecast 6-month revenue** for 25,432 products across 12 regions, achieving 94% accuracy and enabling optimized inventory planning.
- Identified top 20% of customers contributing 62% of total revenue and top-performing products; developed interactive **Power BI** and **Tableau** dashboards visualizing revenue trends, seasonal patterns, and key KPIs, enabled data-driven pricing and promotion strategies.

EXTRACURRICULAR ACTIVITIES:

Research Volunteer (Montclair State University)

November 2024

- Processed NJ weather data (1915-2024) and contributed to research showing ARIMA, SARIMA models outperform deep learning methods in time and space efficiency for **time-series forecasting**.

CERTIFICATIONS:

- Data Science Fundamentals for Data Analysts by Databricks ([Certificate](#))
- Microsoft 365 Fundamentals ([Certificate](#))
- Machine Learning with Python by IBM ([Certificate](#))
- Introduction to Applied Business Analytics by University of Illinois ([Certificate](#))